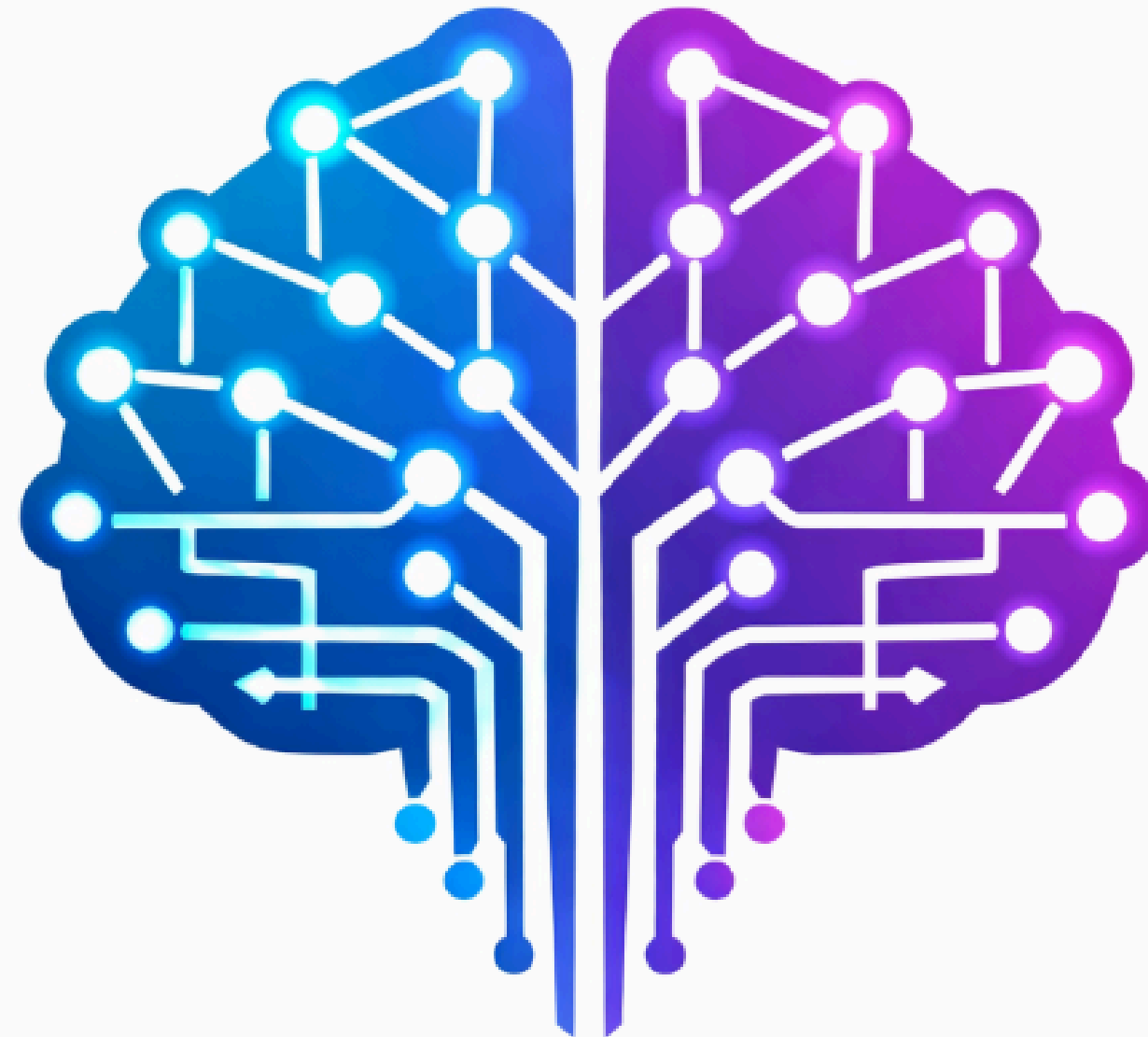


GENAI FUNDAMENTALS



LECTURE:

WHAT MAKES AN LLM POWERFUL (MODEL ANATOMY)

LLMs Are Not Just “Big Chatbots”

Modern LLMs combine multiple ingredients:

- Massive training data
- Large neural networks
- Transformer architecture
- Extensive training compute
- Alignment and safety training

👉 These factors together create powerful AI systems.

Training Data: The Knowledge Source

LLMs learn patterns from massive datasets including:

- books
- websites
- research papers
- conversations
- code repositories

Training data provides the knowledge foundation of the model.

Model Size and Parameters

☞ LLMs contain **billions or trillions of parameters**.

Parameters are numerical weights that store learned patterns.

Larger models generally:

- capture more complex patterns
- understand context better
- produce more coherent responses

Why Different Models Behave Differently

👉 LLM behavior varies because of differences in:

- training datasets
- model architecture
- alignment techniques
- fine-tuning methods

Examples:

- GPT models
- Claude
- Gemini
- LLaMA

👉 Each model is optimized for different goals.

Why Hallucinations Happen

LLMs sometimes generate **incorrect or fabricated information**. This is called **hallucination**.

It happens because the model:

- predicts likely tokens
- does not verify facts
- lacks real-time knowledge

👉 The model prioritizes **coherent language**, not guaranteed correctness.

Why Alignment Matters

Alignment training helps models behave safely and helpfully.

This includes techniques like:

- human feedback
- safety fine-tuning
- reinforcement learning

Alignment helps models:

- follow instructions
- avoid harmful outputs
- behave more responsibly

